

Diffusion Processes for Inference of Structural Causal Models

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Inference of Structural Causal Models

The goal is to infer an SCM-model with optimal

- 1) complexity $\mathcal{L}_C$,
- 2) accuracy $\mathcal{L}_D$,
- 3) robustness $\mathcal{L}_{ci}$

The models:

- 1) the model f_{ai} maps a dataset $\mathbf{D}$ to a model $f_G(\mathbf{w}, \mathbf{x})$,
- 2) the model f_G maps a sample $\mathbf{x} \in \mathbf{D}$ to its forecast $\hat{\mathbf{x}}$, provided optimal parameters $\hat{\mathbf{w}}$.

The graph rewriting

A graph $G(V, E)$ with the

- 1) coloring function $c(V)$,
- 2) weights $\mathbf{w}(V)$,
- 3) probability of the vertices $p(V)$,
- 4) probability of the edges $\mathbf{Z} \in \{0, 1\}^{n \times n}$ or $\mathbf{\Gamma} \in [0, 1]^{n \times n}$.

The graph G defined the function $f_G(\mathbf{w}, \mathbf{x}) \rightarrow \hat{\mathbf{x}}$.

There given a graph rewriting rules $G \mapsto G'$ so that each step delivers an admissible graph (admissible means a graph G defines a function f_G).

To produce a sequence of the graphs introduce a vector $\mathbf{z} = \mathbf{vec}(\mathbf{Z})$.

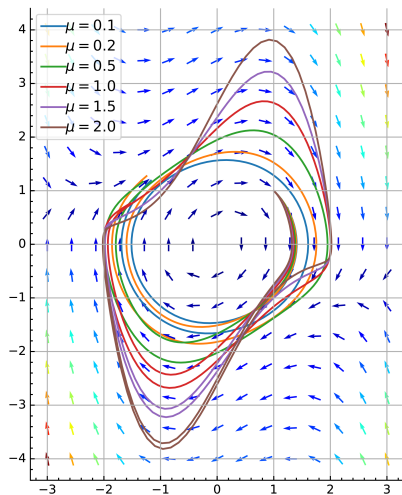
The graph probability

Introduce a Metropolis Hastings graph sampling procedure

The graph probability

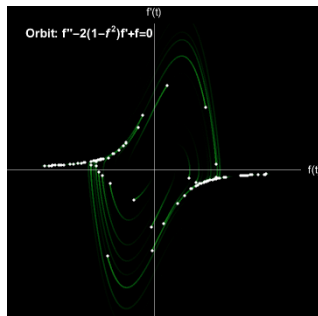
Introduce a Metropolis Hastings graph sampling procedure

Van der Pol oscillator¹

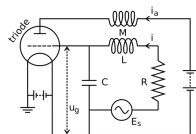


Phase portrait of van der Pol's equation $\frac{d^2 y}{dt^2} + \mu(y^2 - 1) \frac{dy}{dt} + y = 0$.

¹Wikipedia: Van der Pol oscillator

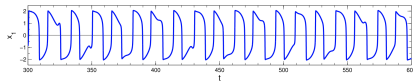


Randomly chosen initial conditions are attracted to a stable orbit



Electrical circuit resulting in a forced Van der Pol oscillator

The van der Pol oscillator model



Chaotic behaviour in the Van der Pol oscillator with sinusoidal forcing.

The van der Pol oscillator model

The forced Van der Pol oscillator:

$$\frac{d^2x}{dt^2} - \mu(1 - x^2)\frac{dx}{dt} + x - A\sin(\omega t) = 0,$$

has two linear and one non-linear parameters: μ, A, ω , where A is the amplitude, of the wave function and ω is its angular velocity.

The causal graph has four vertices, the variables:

$$X_1, X_2, X_3 = \ddot{x}, \dot{x}, x$$

and the time t .

An alternative form is a time-independent Hamiltonian by augmenting it to a four-dimensional, $X_1, \dots, X_4$ autonomous dynamical system using an auxiliary second-order nonlinear differential equation:

$$\ddot{x} - \mu(1 - x^2)\dot{x} + x = 0, \quad \ddot{y} + \mu(1 - x^2)\dot{y} + y = 0.$$

The Hamiltonian

$$H(x, y, p_x, p_y) = p_x p_y + xy - \mu(1 - x^2) y p_y,$$

where

$$p_x = \dot{y} + \mu(1 - x^2)y \quad \text{and} \quad p_y = \dot{x}$$

are the conjugate momenta corresponding to "x" and "y".

Slide 2

1. Goals
2. Requirements to model
3. How to construct a model
4. Algebraical and Statistical assumptions
5. Do-intervention and testing (testability)
6. Local and global modeling expand
7. Graph diffusion and reconstruction
8. Neural ODE is possible
9. Fine-tuning
10. And testing
11. The general probabilistic model and loss function
12. Non-parametric modeling and Optimal sampling
13. Time and space segmentation